SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

COLLECTIONS EXERCISE

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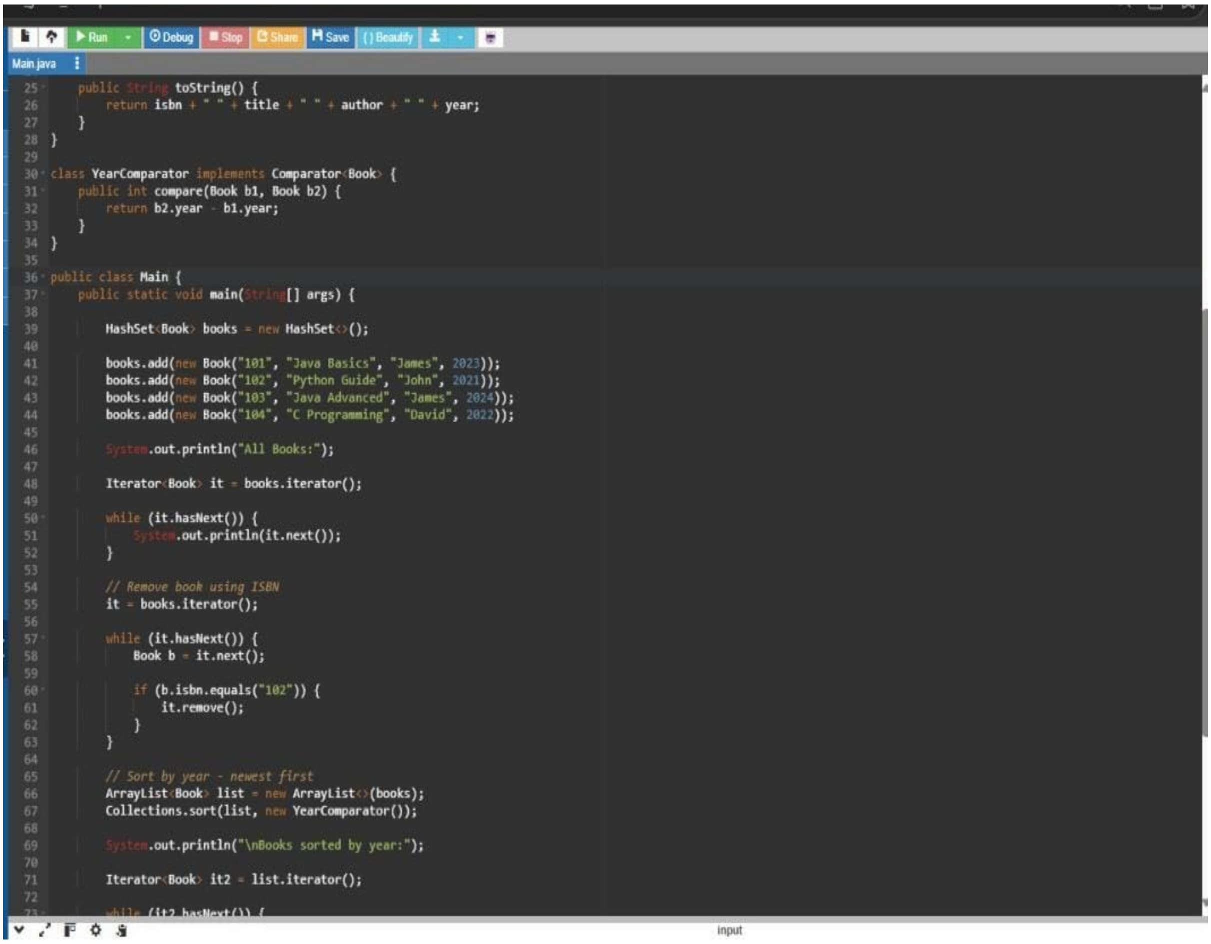
COURSE:PROGRAMMING IN JAVA

COURSE CODE:CSA0904

1. Library Book Management System

What it uses

- HashSet<Book> → avoids duplicate ISBNs
- Iterator → display and remove books
- Comparator → sort by publication year
- Search books by author
-



```java
    public String toString() {
        return isbn + " " + title + " " + author + " " + year;
    }
}

class YearComparator implements Comparator<Book> {
    public int compare(Book b1, Book b2) {
        return b2.year - b1.year;
    }
}

public class Main {
    public static void main(String[] args) {

        HashSet<Book> books = new HashSet<>();

        books.add(new Book("101", "Java Basics", "James", 2023));
        books.add(new Book("102", "Python Guide", "John", 2021));
        books.add(new Book("103", "Java Advanced", "James", 2024));
        books.add(new Book("104", "C Programming", "David", 2022));

        System.out.println("All Books:");

        Iterator<Book> it = books.iterator();

        while (it.hasNext()) {
            System.out.println(it.next());
        }

        // Remove book using ISBN
        it = books.iterator();

        while (it.hasNext()) {
            Book b = it.next();

            if (b.isbn.equals("102")) {
                it.remove();
            }
        }

        // Sort by year - newest first
        ArrayList<Book> list = new ArrayList<>(books);
        Collections.sort(list, new YearComparator());

        System.out.println("\nBooks sorted by year:");

        Iterator<Book> it2 = list.iterator();

        while (it2.hasNext()) {
```

```
All Books:
101 Java Basics James 2023
102 Python Guide John 2021
103 Java Advanced James 2024
104 C Programming David 2022

Books sorted by year:
103 Java Advanced James 2024
101 Java Basics James 2023
104 C Programming David 2022

Books by James:
101 Java Basics James 2023
103 Java Advanced James 2024
```

2. Student Grade Analyzer

What it uses

- ArrayList<Student>
- HashMap<String,Integer>
- Iterator
- Collections.sort()
- Custom Comparator
- Average calculation
- Top 3 students

- Removing students who failed more than 2 subjects

Main.java

```java
class Student {
    int id;
    String name;
    HashMap<String, Integer> grades;

    Student(int id, String name) {
        this.id = id;
        this.name = name;
        grades = new HashMap<>();
    }

    double average() {
        int total = 0;

        for (int mark : grades.values()) {
            total = total + mark;
        }

        return (double) total / grades.size();
    }

    int failedSubjects() {
        int count = 0;

        for (int mark : grades.values()) {
            if (mark < 40) {
                count++;
            }
        }

        return count;
```

input

```
Student Averages:
Kaviya = 85.0
Arun = 70.0
Priya = 92.33333333333333
Rahul = 30.0

Top 3 Students:
Priya - Average: 92.33333333333333
Kaviya - Average: 85.0
Arun - Average: 70.0
```

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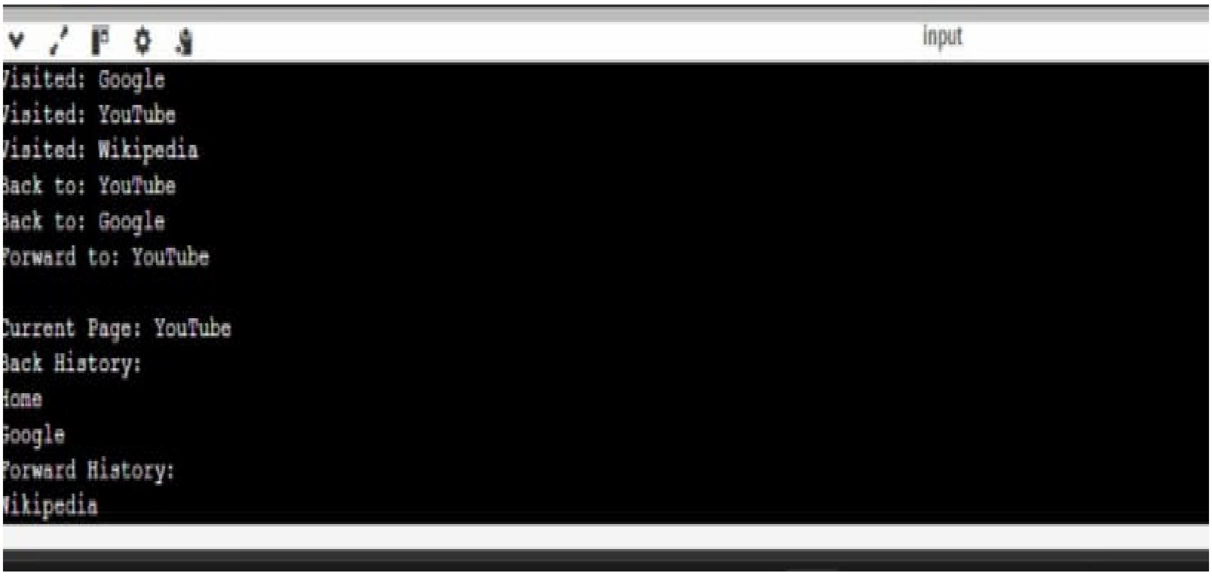
3. Navigation History – Web Browser

What it uses

- Stack<String> for back history
- Stack<String> for forward history
- Iterator
- visitPage()
- goBack()
- goForward()

```java
import java.util.*;

class Browser {

    Stack<String> back = new Stack<>();
    Stack<String> forward = new Stack<>();

    String current = "Home";

    void visitPage(String url) {

        back.push(current);
        current = url;

        forward.clear();

        System.out.println("Visited: " + current);
    }

    void goBack() {

        if (back.empty()) {
            System.out.println("No page to go back");
            return;
        }

        forward.push(current);
        current = back.pop();

        System.out.println("Back to: " + current);
    }

    void goForward() {

        if (forward.empty()) {
            System.out.println("No page to go forward");
            return;
        }
```



4. Employee Payroll System

What it uses

- HashMap<String, List<Employee>>
- ArrayList<Employee>
- Iterator
- Safe removal using Iterator.remove()
- Department-wise salary
- Highest-paid employee
- Moving employee

```java
    }

    public String toString() {
        return id + " " + name + " " + salary;
    }
}

public class Main {

    static HashMap<String, List<Employee>> employees = new HashMap<>();

    static void addEmployee(Employee e) {

        if (!employees.containsKey(e.department)) {
            employees.put(e.department, new ArrayList<>());
        }

        employees.get(e.department).add(e);
    }

    static void display() {

        Iterator<Map.Entry<String, List<Employee>>> it =
                employees.entrySet().iterator();

        while (it.hasNext()) {

            Map.Entry<String, List<Employee>> entry = it.next();

            System.out.println("\nDepartment: " + entry.getKey());

            Iterator<Employee> eit = entry.getValue().iterator();

            while (eit.hasNext()) {
                System.out.println(eit.next());
            }
        }
    }

    static void departmentSalary() {

        System.out.println("\nTotal Salary:");

        Iterator<Map.Entry<String, List<Employee>>> it =
                employees.entrySet().iterator();

        while (it.hasNext()) {

            Map.Entry<String, List<Employee>> entry = it.next();

            double total = 0;

            Iterator<Employee> eit = entry.getValue().iterator();

            while (eit.hasNext()) {
                total = total + eit.next().salary;
            }

            System.out.println(entry.getKey() + " = " + total);
        }
    }
```

```
ighest Paid Employee:
R = Meena
T = Priya

fter removing salary below 55000:

epartment: HR
104 Meena 80000.0

epartment: IT
101 Arun 60000.0
102 Priya 75000.0
run moved from IT to HR

fter moving employee:

epartment: HR
104 Meena 80000.0
101 Arun 60000.0
```